

Who will buy again

First purchase is only the start. Repeat buyer rate is 7.05% (255 of 3,615). Rank products on long-term LTV, not day-one sales. Instead of prioritizing only footwear products focus can be shifted to sports and beauty products.

Target: is_repeat_buyer

Dataset Overview

DATASET NAME	ROWS	COLUMNS
GA4 E-Commerce Shopper Intelligence.csv	3615	22

Key Performance Indicators

REPEAT BUYER PREDICTION 0.89 AUC	CORRECT CUSTOMER CLASSIFICATION 97.6% Accuracy	OBSERVED REPEAT BUYER RATE 7.05% 255 repeat buyers in the build	TOP REPEAT-BUYER CATEGORY Sports & Fitness LTV Rank #1 vs Day-1 Rank #3
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Current DXI · Repeat Buyer Target

CURRENT DXI

3.88

REPEAT BUYERS

7.05%

REPEAT BUYERS

255

NON REPEAT BUYERS

3360

TARGET IMPROVEMENT (BUILD)

+20% Repeat buyers

Build target improvement is +20% (increasing) on is_repeat_buyer — Repeat buyers · after 3 months.

CURRENT REPEAT BUYERS

7.05%

TARGET AFTER +20%

8.46%



Current vs Target DXI

CURRENT DXI

3.88

CURRENT REPEAT BUYERS

7.05%

MODEL ACCURACY

97.6%

TARGET DXI

3.44

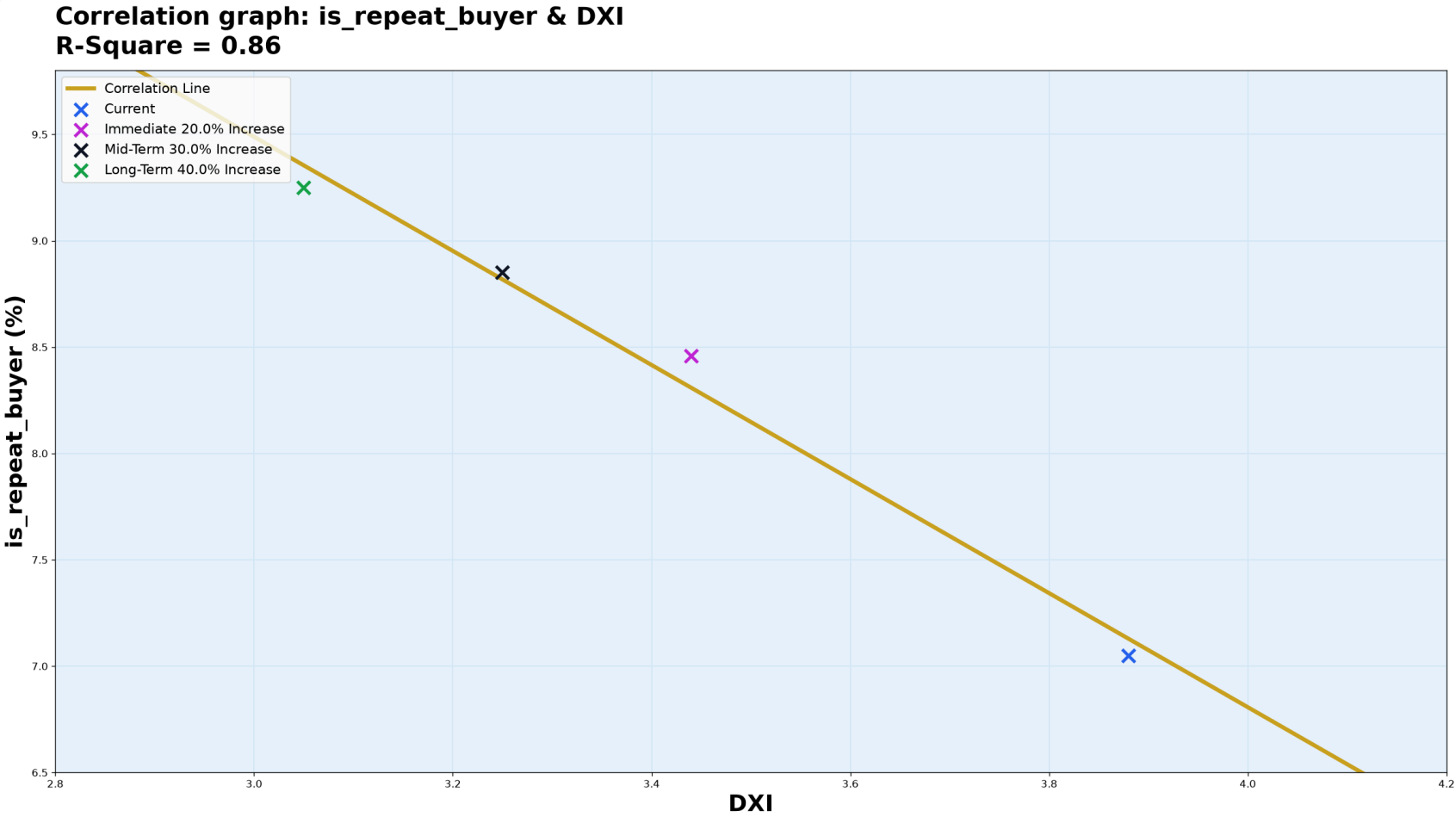
TARGET REPEAT BUYERS

8.46%

REPEAT BUYERS CHANGE

+20%

Correlation Chart



3 Levels of Improvement

Immediate

Target DXI **3.44**

Mid-term

Target DXI **3.25**

Long-term

Target DXI **3.05**

Projected outcome8.46%

Improvement+20%

Projected outcome9.09%

Improvement+29%

Projected outcome9.73%

Improvement+38%



Repeat Buyer Rank Movement Highlights

TOP LTV LEADER

Sports & Fitness

+2 Jump

Day-1 Rank #3 → LTV Rank #1

BIGGEST CLIMBER

Beauty

+2 Jump

Day-1 Rank #6 → LTV Rank #4

LOWEST LONG-TERM VALUE

Footwear

-4 Jump

Day-1 Rank #2 → LTV Rank #6

Repeat Buyer Drivers — Target Top 5

#	Feature	Contribution %
1	Bounce Rate	56.0%
2	Items Added To Cart	11.1%
3	Engagement Time	7.7%
4	Order Month	9.2%

Category Action Tiers & Strategy

Tier 1: LTV Champions (Score 8+)

Sports & Fitness

Undervalued on day-one sales but generate top repeat retention.

Action: Reallocate performance ad spend here.

Tier 2: Strong Performers (Score 6–8)

Electronics

Solid repeat rates with room to scale bundles.

#	Feature	Contribution %
5	Churn Risk	4.0%

Repeat Buyer Drivers: Minimizing Bounce Rate accounts for 56% of repeat-buyer conversions — immediate site engagement is the biggest factor in bringing customers back.

Category focus: Shift priority from footwear-only toward Sports & Beauty for long-term LTV, not day-one sales.

Action: Implement "frequently bought together" bundles.

Tier 3 & 4: Needs Strategy (Score < 6)

Accessories, Beauty, Apparel, Footwear

Overvalued on initial sales but suffer high single-purchase drop-off.

Action: Trigger win-back email sequences within 14 days.

Product Category Repeat Buyer Leaderboard

Rank	Product Category	Composite LTV Score	Avg LTV (\$)	Repeat Buyer Rate (%)	Avg Churn Risk (%)	Strategic Action Tier
1	Sports & Fitness	9.4	\$4120	11.2%	0.28%	Tier 1: LTV Champion
2	Electronics	7.2	\$3890	9.1%	0.36%	Tier 2: Strong Performer
3	Accessories	4.8	\$3650	7.4%	0.48%	Tier 2: Strong Performer
4	Beauty	3.1	\$3480	6.8%	0.55%	Tier 4: Needs Strategy
5	Apparel	2.4	\$3310	5.9%	0.62%	Tier 4: Needs Strategy
6	Footwear	1.5	\$3050	4.8%	0.71%	Tier 4: Needs Strategy

 Model Performance

Accuracy	97.6%
AUC	0.89
Precision	86.4%
Recall	79.1%
F1	82.6%

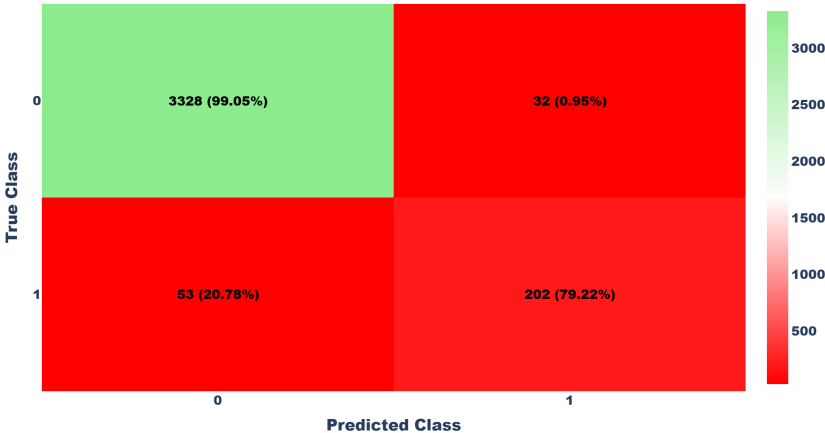
 Current Business Status

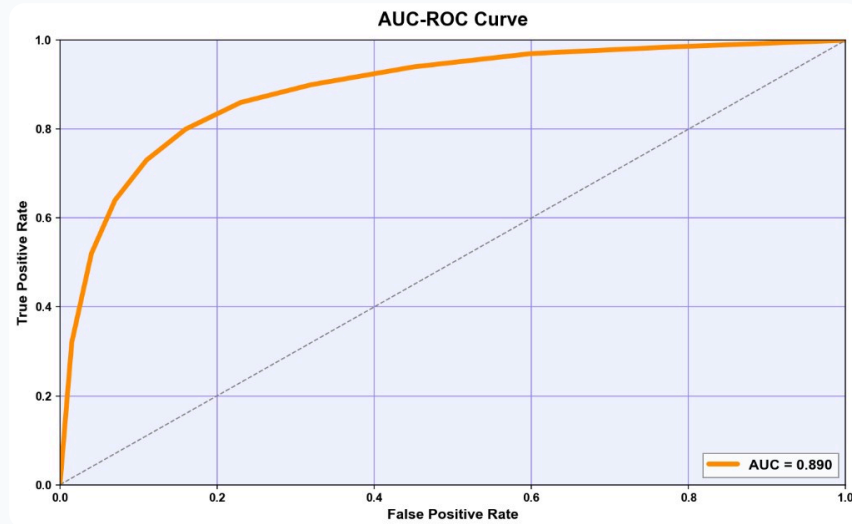
 Current DXI	3.88
 Repeat buyers	7.05%
 After 3 months	8.46% · +20%
 Business lens	Shift focus from footwear to Sports & Beauty; minimize bounce.

 AUC / ROC

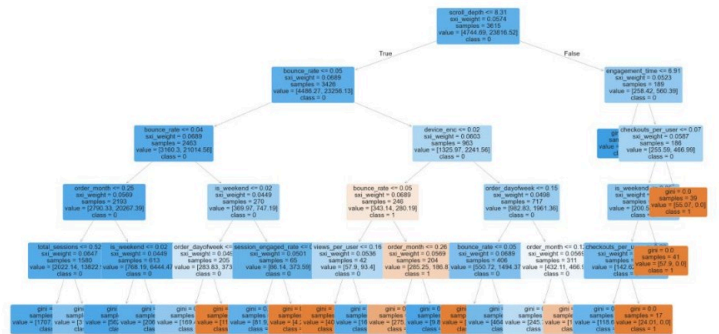
 Confusion Matrix

Accuracy Matrix





📌 Target Decision Tree — Business Interpretation



- **1st node:** Bounce Rate (root driver) — accounts for 56% of repeat-buyer conversions.

🚩 Feature Importance — Target Top 5

#	Feature	Importance	Contribution %
1	Bounce Rate	55.96	56.0%
2	Items Added To Cart	11.10	11.1%
3	Engagement Time	7.70	7.7%
4	Device ENC	5.10	5.1%
5	Churn Risk	4.00	4.0%

- Immediate site engagement is the single biggest factor in bringing customers back.
- Focus targeting on low-bounce users with active carts to convert returning visitors into repeat buyers.
- Top contributions: Bounce Rate 56.0%, Items Added To Cart 11.1%, Engagement Time 7.7%.

Business actions

- First purchase is only the start — repeat rate is 7.05% (255 of 3,615); lift to 8.46% (+20%) after 3 months.
- Instead of prioritizing only footwear, shift focus to Sports and Beauty for long-term LTV.
- Minimize Bounce Rate (56% contribution); target low-bounce users with active carts.
- Model: Accuracy 97.6% · AUC 0.89 · Precision 86.4% · Recall 79.1%.
- Correlation R^2 0.86 · Current DXI 3.88 · short-term DXI 3.44.